



Mechanics of displacive instabilities in solids

Habilitation à Diriger des Recherches (HDR)

Oğuz Umut Salman

Laboratoire des Sciences des Procédés et des Matériaux (LSPM)
CNRS UPR 3407 | Université Sorbonne Paris Nord

31 Mars 2026

Rapporteurs

Mme Maurine MONTAGNAT
M. Muhittin MUNGAN
M. Aurélien VATTRÉ

DR CNRS, IGE
Professeur, Univ. of Cologne
MR Onera, Univ. Paris-Saclay

Examineurs

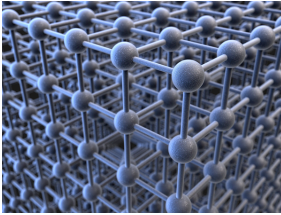
Mme Véronique LAZARUS
M. Nicolas AUFRAY
M. Denis GRATIAS
M. Damien FAURIE
M. Ioan R. IOANESCU

Professeure, ENSTA Paris
Professeur, Sorbonne Université
DR émérite CNRS, IRCP
Professeur, USPN
Professeur, USPN

Non-linear phenomena

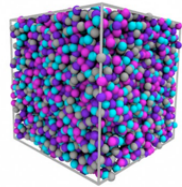
Crystal Transitions and Disorder

Crystalline



Stronger disorder
→
←
Weak disorder

Amorphous

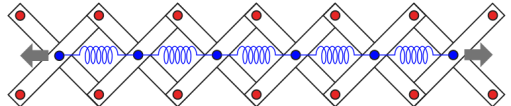


Meta-materials

spring-chain



spring-chain system embedded into a pantograph



Scientific Context: Nonlinearity & Instability

Scientific Challenges

The Challenge:

Understanding material behavior beyond the elastic limit where microscale rearrangements drive macroscopic nonlinearity.

Key Phenomena:

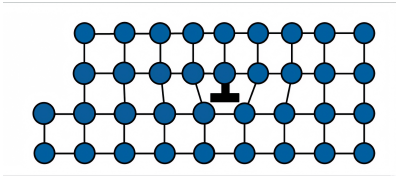
- ▶ **Phase Transitions:** Martensitic transformations.
- ▶ **Plasticity:** Dislocation-mediated flow and avalanches.
- ▶ **Fracture:** Brittle-to-ductile transitions and damage.

Unifying Perspective

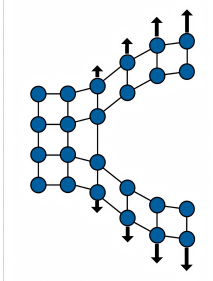
These phenomena are inherently **displacement-driven** (*non-diffusive*) and can be studied within the **same context** of displacive instabilities.

Common Physical Mechanisms: Displacive Instabilities

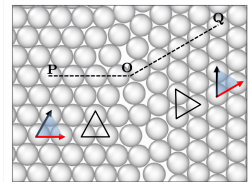
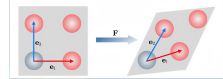
Plasticity



Fracture



Phase Transition

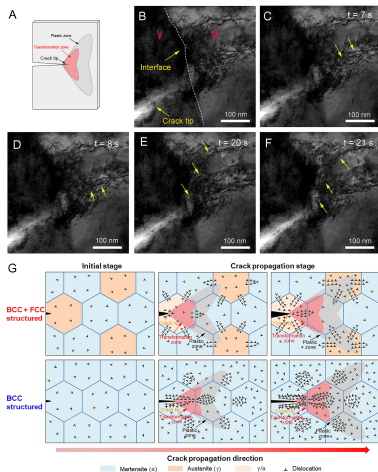


Central Theme

Uniform treatment through the lens of non-convex energy and geometric frustration.

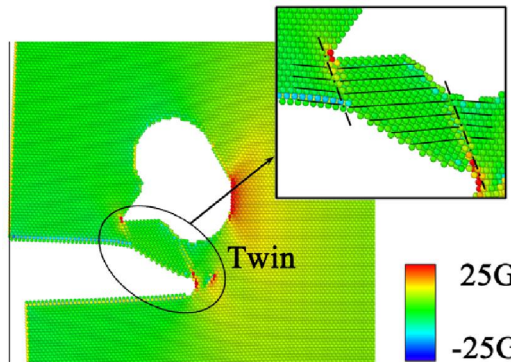
Motivation for Unification

Experimental Observations



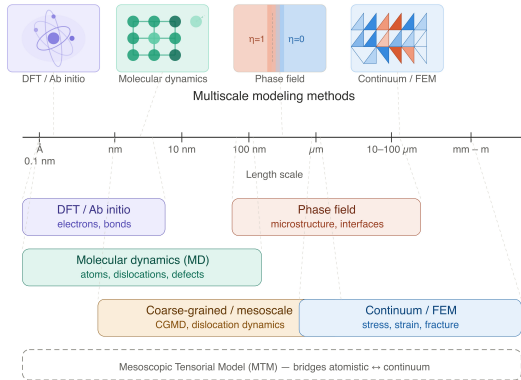
Zhang et al. PNAS, 2025

Computational Modeling



Liu, Groh, Mater. Sci. Eng., 2014

Multiscale Modeling Framework



Integrating information from the discrete lattice to the continuous engineering scale.